Cranial nerves 1, 3, 4, & 6

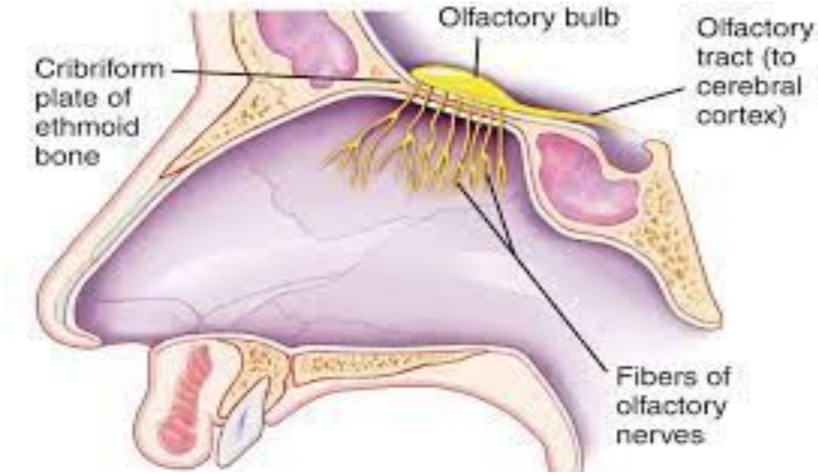
Learning Objectives:

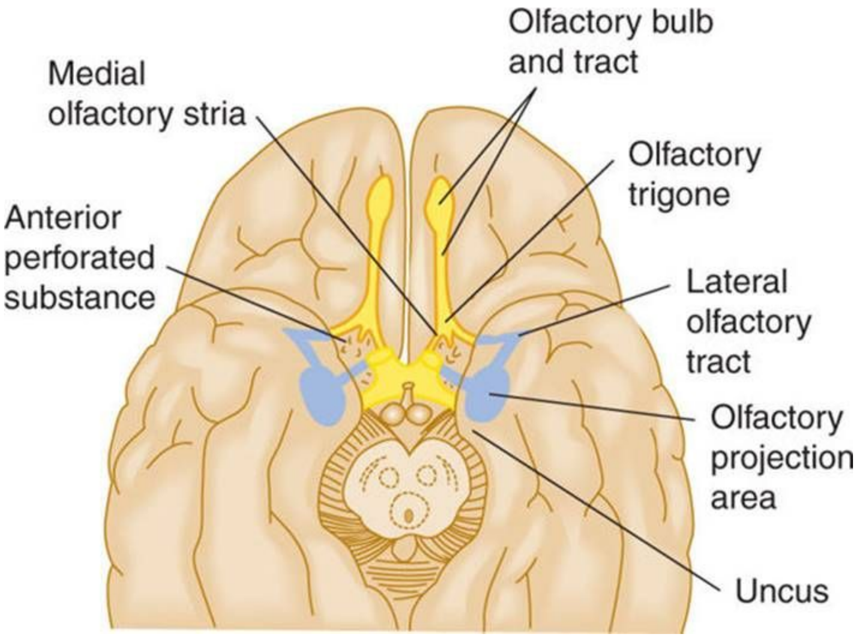
At the end of the lecture, the students should be able to:

- **Enlist the main contents of the orbit.**
- **Define the orbital and bulbar fascia.**
- **Describe the basic course and distribution of the 3^{rd}, 4^{th}, & 6^{th} cranial nerves.**
- **Define the ciliary ganglion and its connections.**
- **Describe the pathway of Pupillary(light) reflex.**
- **Describe the main blood supply of the orbit.**

Olfactory nerve (1st):

- It is sensory in nature.
- It arises from neuroepithelium of upper one-third of nasal cavity.
- This neuroepithelium contains bipolar neurons.
- 18-20 olfactory nerve fibers arise from these bipolar neurons and pass through cribriform plate of ethmoid bone.
- These olfactory nerve fibers synapse with the cells of olfactory bulb.
- The olfactory tract travels posteriorly on the inferior surface of frontal lobe and ends in the olfactory trigone.
- The olfactory trigone is a triangular widening of the terminal olfactory tract located superior to the anterior clinoid process and to the anterior perforated substance.





Olfactory pathways- Comprises

Olfactory nerve- bipolar neuron pierces to cribriform palate to reach olfactory bulb

Olfactory bulb- contains mitral cell, tufted cell, periglomerular cell

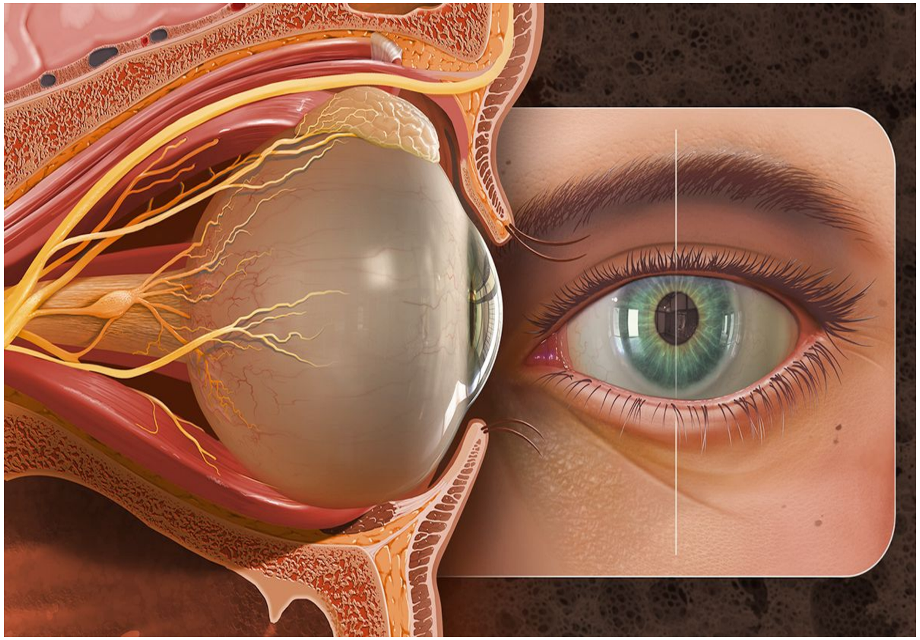
Olfactory tract - neurons recieve synaptic connections from olfactory bulb which excite inhibitory contralateral olfactory bulb.

Olfactory cortex- includes anterior olfactory nucleus ,prepiriform cortex, olfactory tubercle,and amygdala(parts if limbic system)

Contents of the Orbit:

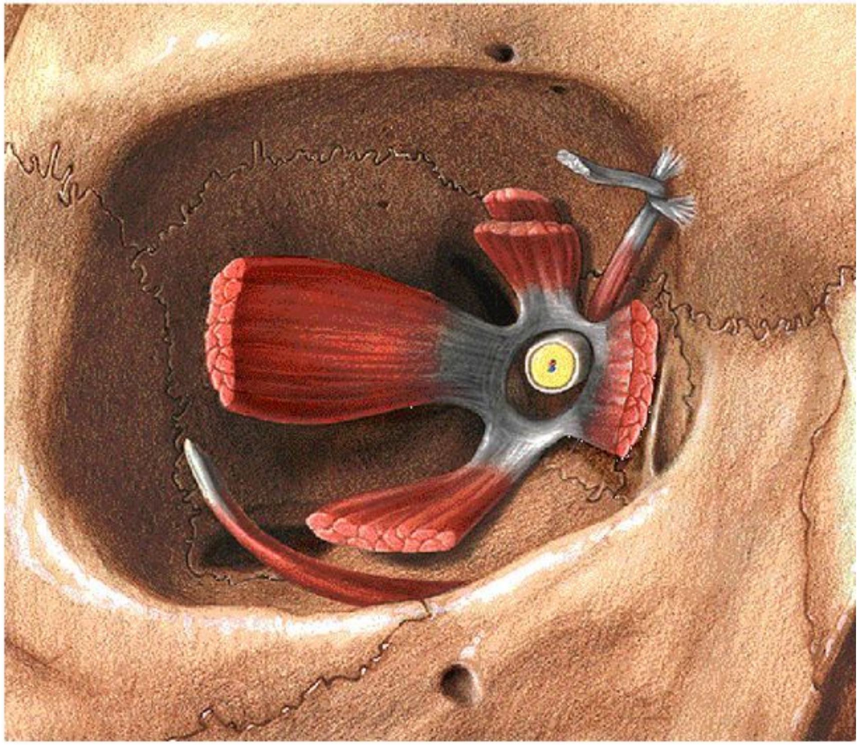
The main contents of the orbit are listed below:

- **Eyeball**
- **Orbital & Bulbar Fascia**
- **Extraocular Muscles**
- **Ophthalmic Artery**
- **Superior & Inferior Ophthalmic Veins**
- **Optic, Occulomotor, Trochlear, & Abducent Nerves; branches of Ophthalmic & maxillary Nerves**
- **Lacrimal Gland**



Contents of the Orbital Cavity

- •Orbital fat
- •Eyeball
- •Muscles
- •Vessels
- •Nerves
- •Ciliary ganglion
- •Lacrimal gland



Muscles of the Orbit

Levator palpebrae suprioris

Extrinsic muscles of the eye

Intrinsic muscles of the eye

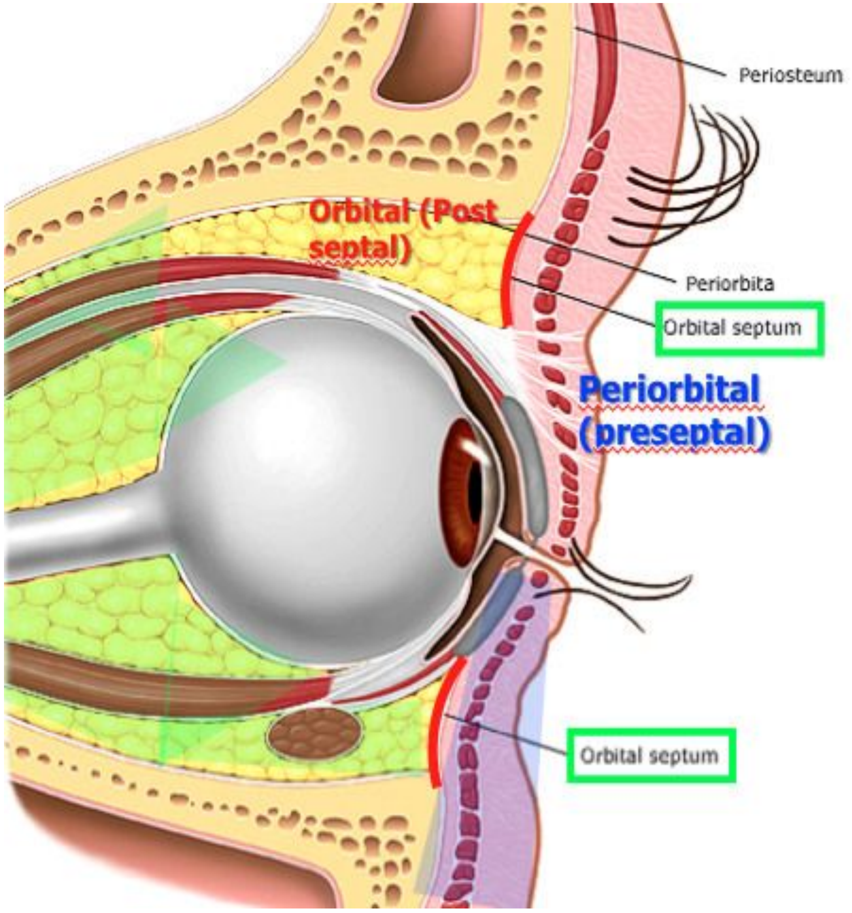
4 Recti muscles
- •Superior rectus
- •Inferior rectus
- •Medial rectus
- •Lateral rectus

2 Oblique muscle
- •Superior oblique
- •Inferior oblique

Intrinsic muscles of the eye:
- •Dilator pupillae
- •Constrictor pupillae
- •Ciliary muscle

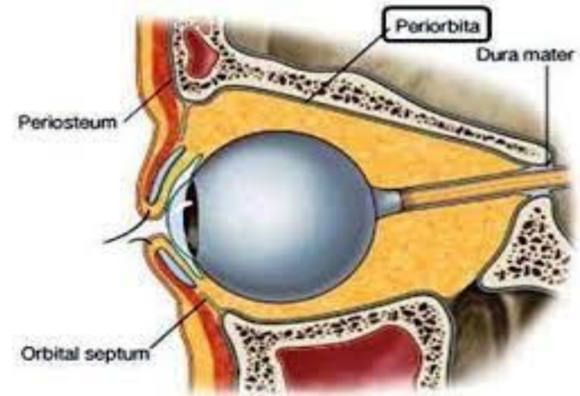
Orbital Fascia/Periorbita:

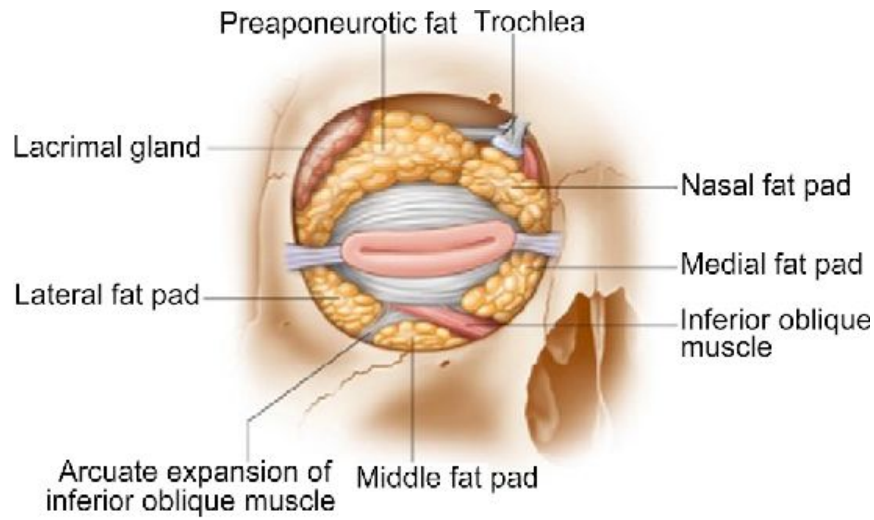
- It forms the periosteum of the bony orbit.
- It is loosely attached to the underlying bone.
- Posteriorly it is continuous with the dura mater and sheath of the optic nerve.
- Anteriorly it is continuous with the periosteum lining the bones around the orbital margin.



Modifications of the Periobita:

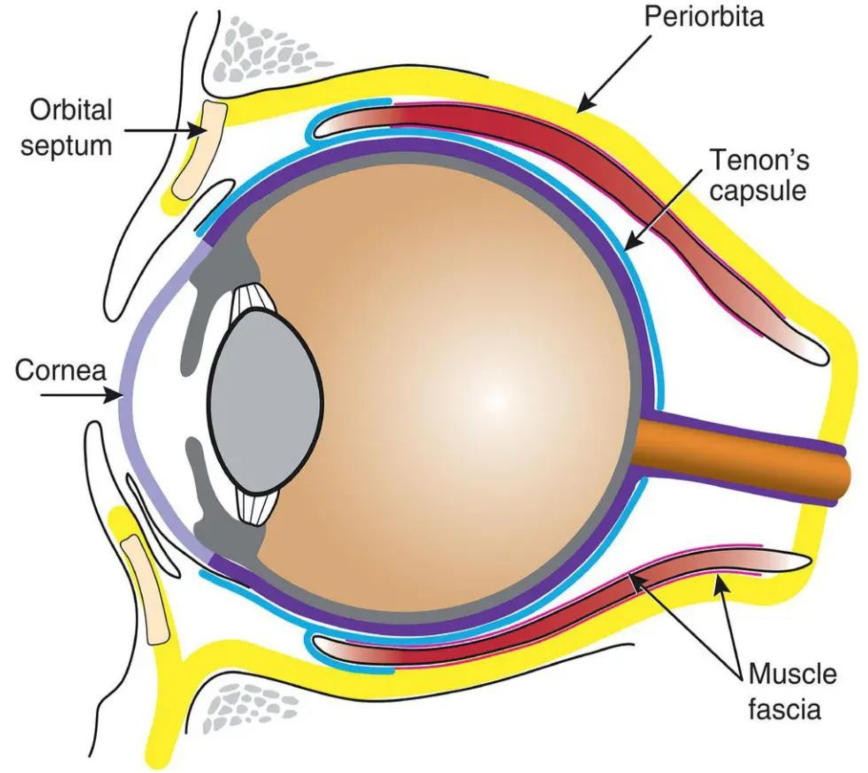
1) At the upper & lower margins of the orbit, it sends off flap-like continuations in the eyelids forming the **Orbital Septum.**
2) **Trochlea:** a fibrous pulley to hold the tendon of the superior oblique muscle.
3) **Lacrimal Fascia:** bridging the lacrimal groove.





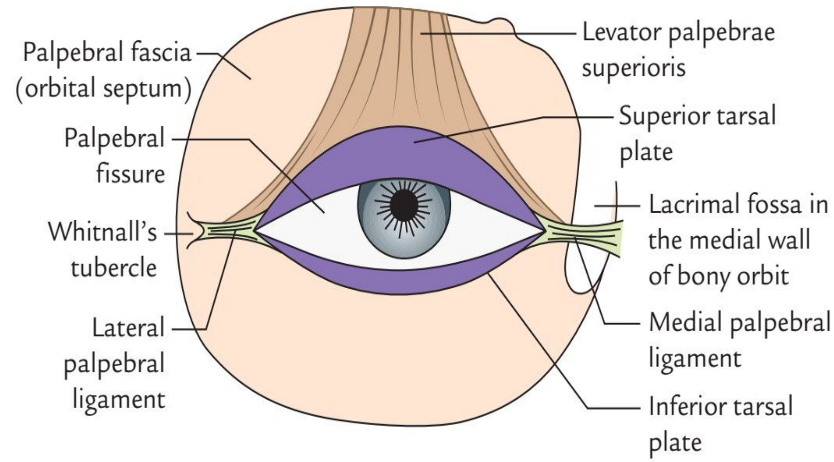
Bulbar Fascia:

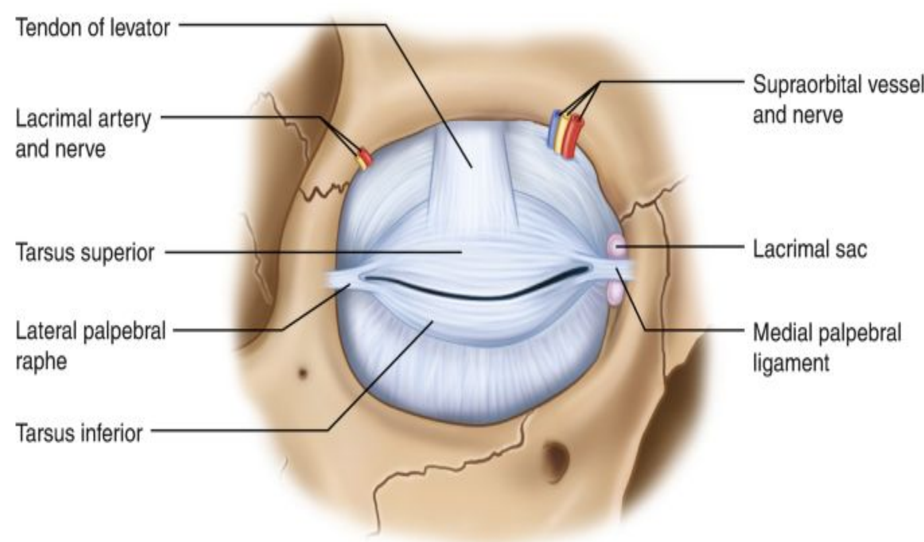
- Also known as Tenon's Capsule.
- It is a thin, loose membrane surrounding the eyeball.
- It is separated from the sclera by the episcleral space.
- It extends from the optic nerve to the sclerocorneal junction (limbus).



Modifications of the Bulbar Fascia:

1) Forms **tubular sheaths** around the tendons of each orbital muscle.
2) **Medial Palpebral (Check) Ligament**
3) **Lateral Palpebral (Check) Ligament**
4) **Suspensory Ligament of the eye (Ligament of Lockwood)**





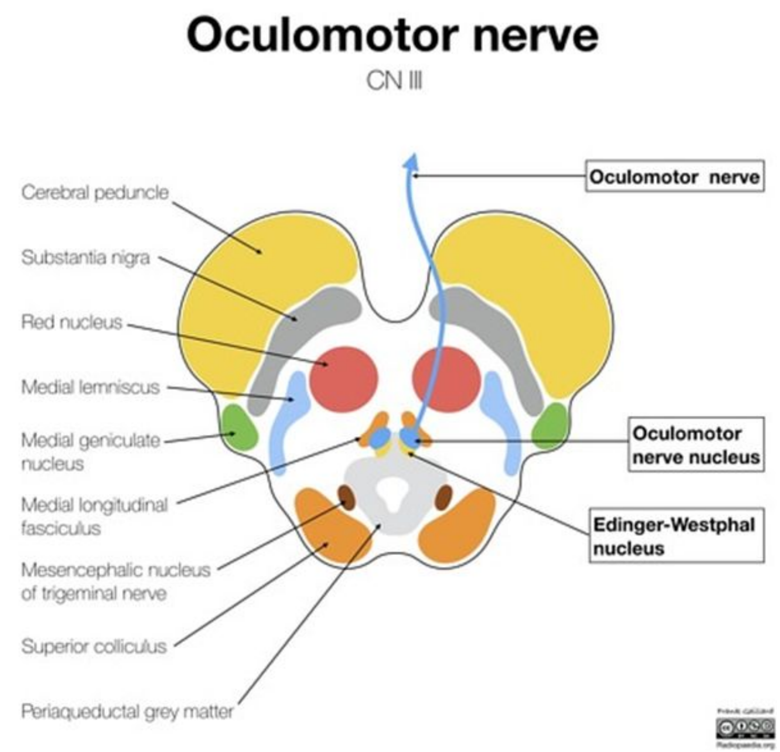
Oculomotor nerve (3^{rd}):

- It is a motor nerve.
- It supplies extraocular as well as intraocular muscles.
- **Nuclei:**

I. Oculomotor nucleus which is located in the central grey matter of midbrain at the level of superior colliculus.

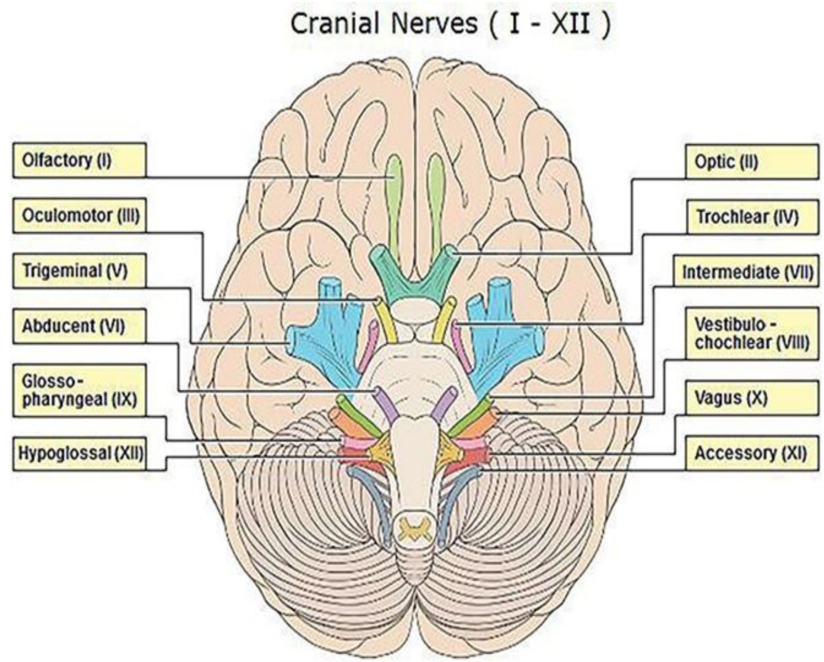
II. Edinger Westphal nucleus is a part of oculomotor nuclear complex from which fibers for constrictor pupillae and ciliary muscles arise.

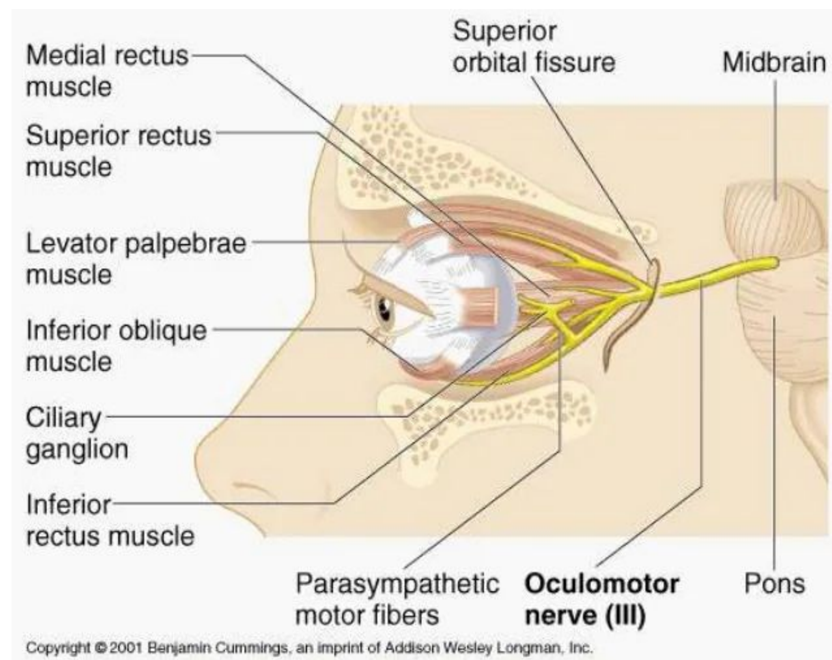
- The oculomotor nucleus is connected to pretectal nuclei for the light reflex.



Course:

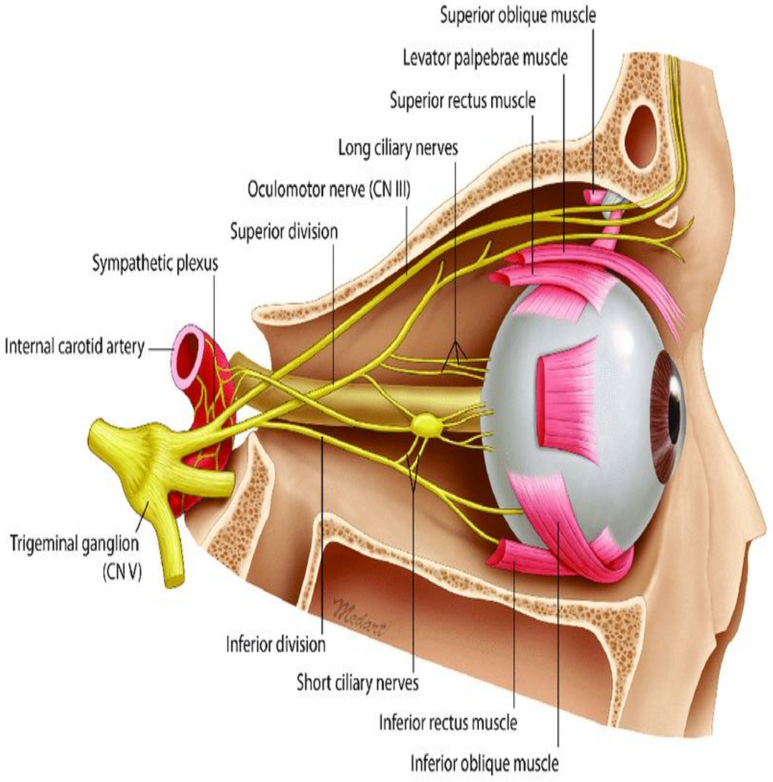
- At the base of brain, the nerve is attached to oculomotor sulcus medial to crus cerebri.
- Then it runs through interpeduncular fossa to reach the cavernous sinus.
- It runs in the lateral wall of cavernous sinus above trochlear nerve.
- It splits into an upper smaller division and a lower larger division in the anterior part of cavernous sinus.
- Then enters the orbit through superior orbital fissure and supplies the extraocular muscles through upper and lower divisions.





CILIARY GANGLION

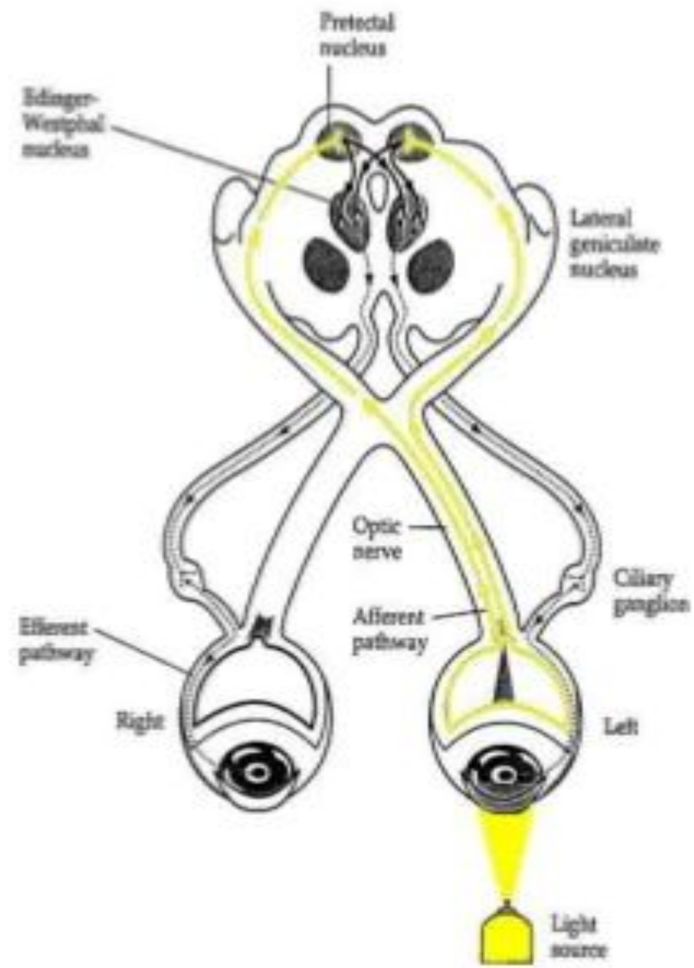
- **SITE:** in the fat **between** optic nerve & lateral rectus muscle
- **NUCLEUS: Edinger Westphal** of occulomotor (midbrain)
- **PREGANGLIONIC FIBERS:** along occulomotor through **nerve to inferior oblique**
- **POSTGANGLIONC FIBERS:** along short ciliary nerves to **sphincter pupillae & ciliary muscles**



Light reflex:

- **Pupillary light reflex (PLR)** : that controls the diameter of the pupil, in response to the intensity of light that falls on the retinal ganglion cells of the retina in the back of the eye.

- **Stimulus:** direct light (torch).
- **Affrent:** optic nerve fibers(**CNII**).
- **Center:** B/S
- **Eefrent:** short ciliary nerve fibers (**CNIII**)
- **Effector organ:** Pupillary constrictor muscle.
- **General pathway (**important**)**
  Ganglion cell → optic nerve → pretectal nuclei→ Edinger-Westphal nucleus→ the ciliary ganglion→ short ciliary nerve fiber→ Pupillary constrictor.



Applied anatomy:

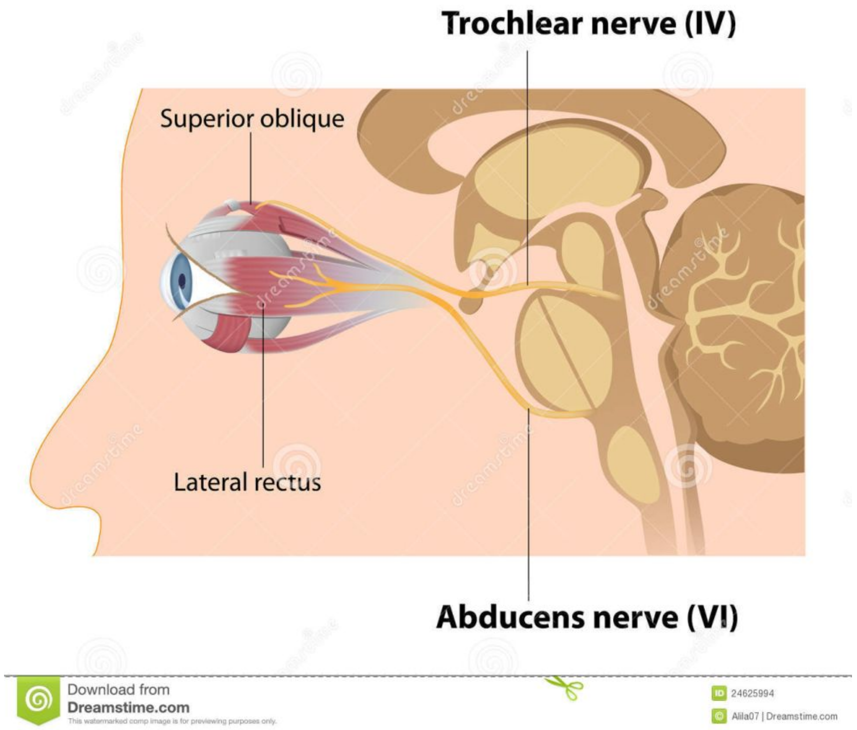
- Complete paralysis of 3^{rd} nerve results in:

I. Ptosis (drooping of upper eyelid)
II. Lateral squint
III. Dilatation of pupil
IV. Loss of accomodation
V. Mild proptosis (forward projection of eyeball)
VI. Diplopia

- **Squint:** it occurs as a result of weakness/paralysis of any extraocular muscle in which both eyeballs are not aligned in the same direction.
- Squint may be congenital or acquired.

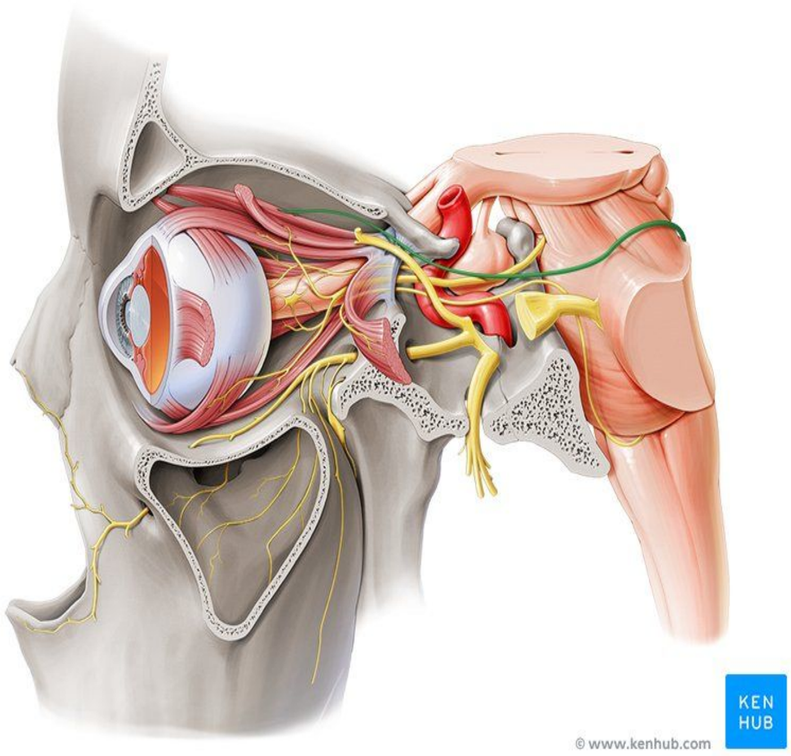
Trochlear nerve (4^{th}):

- It is motor and supplies superior oblique muscle of the eyeball.
- **Nucleus:** trochlear nucleus is located in the central grey matter of midbrain at the level of inferior colliculus.



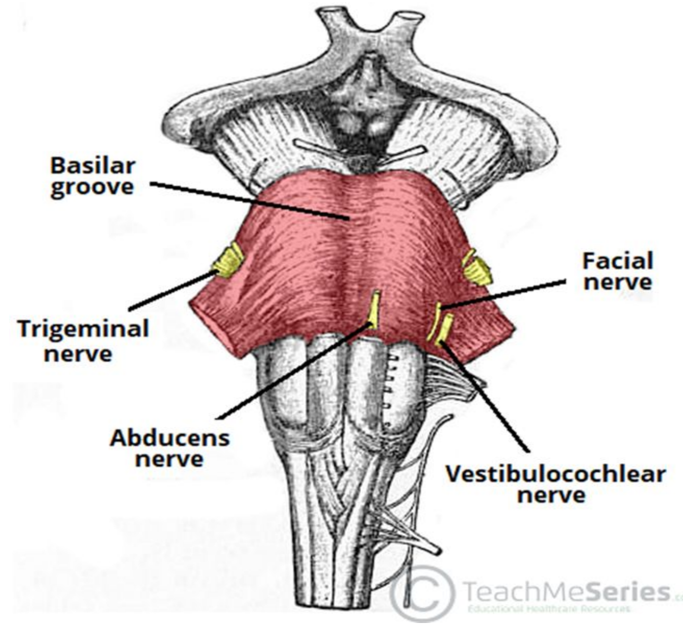
Course:

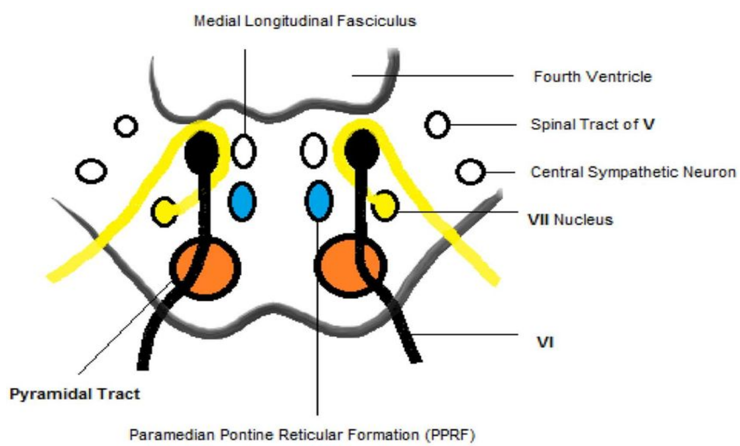
- The trochlear nerve appears from the dorsal region of the brainstem around the level of caudal mesencephalon below the inferior colliculus.
- It is the only nerve which arise from dorsal aspect of brainstem.
- It winds round superior cerebellar peduncle and cerebral peduncles to reach the ventral surface between temporal lobe and upper border of pons.
- Then it enters the cavernous sinus and runs in the lateral wall between oculomotor and ophthalmic nerves.
- Finally it enters the orbit through superior orbital fissure and ends by supplying superior oblique muscle.
- **Applied anatomy:** When trochlear nerve is damaged, *diplopia* occurs on looking downwards.

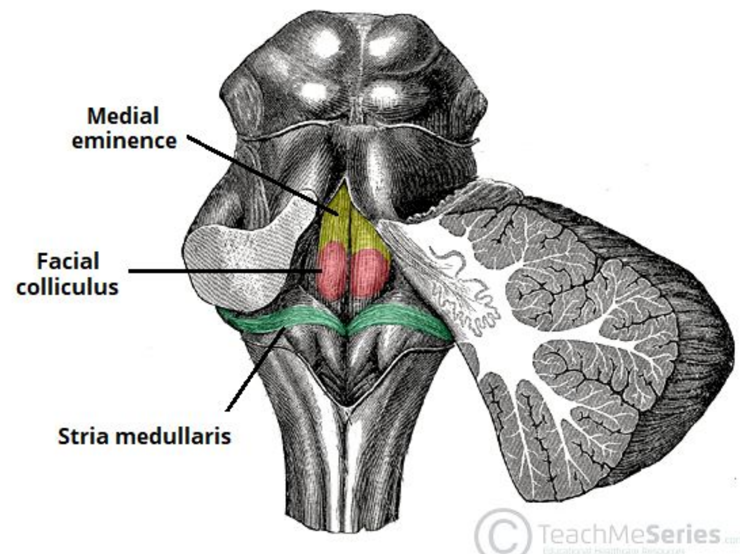


Abducent nerve (6^{th}):

- It is motor and supplies lateral rectus muscle.
- **Nucleus:** abducent nucleus situated in the upper part of floor of fourth ventricle beneath facial colliculus.



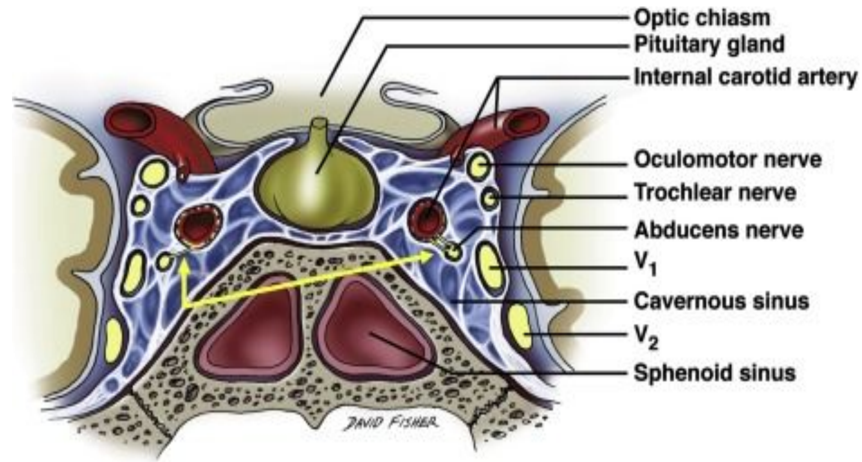


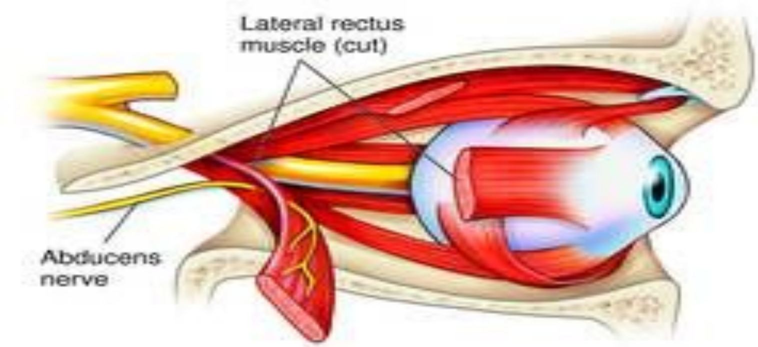


Course:

- The nerve is attached to lower border of pons opposite upper end of pyramid.
- It enters the cavernous piercing its roof and passes through its center lateral to internal carotid artery.
- It enters orbit passing through superior orbital fissure and ends by supplying only lateral rectus muscle.
- **Applied anatomy:** Damage to abducent nerve results in:

a) Internal/medial squint, &
b) Diplopia

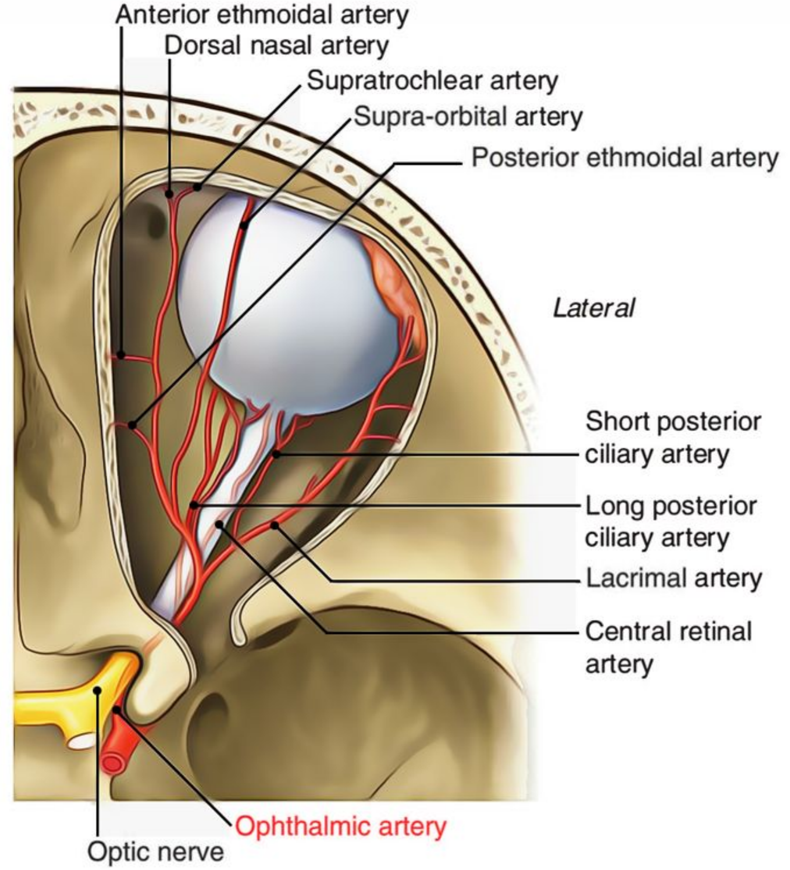




Ophthalmic Artery:

Origin & Course:

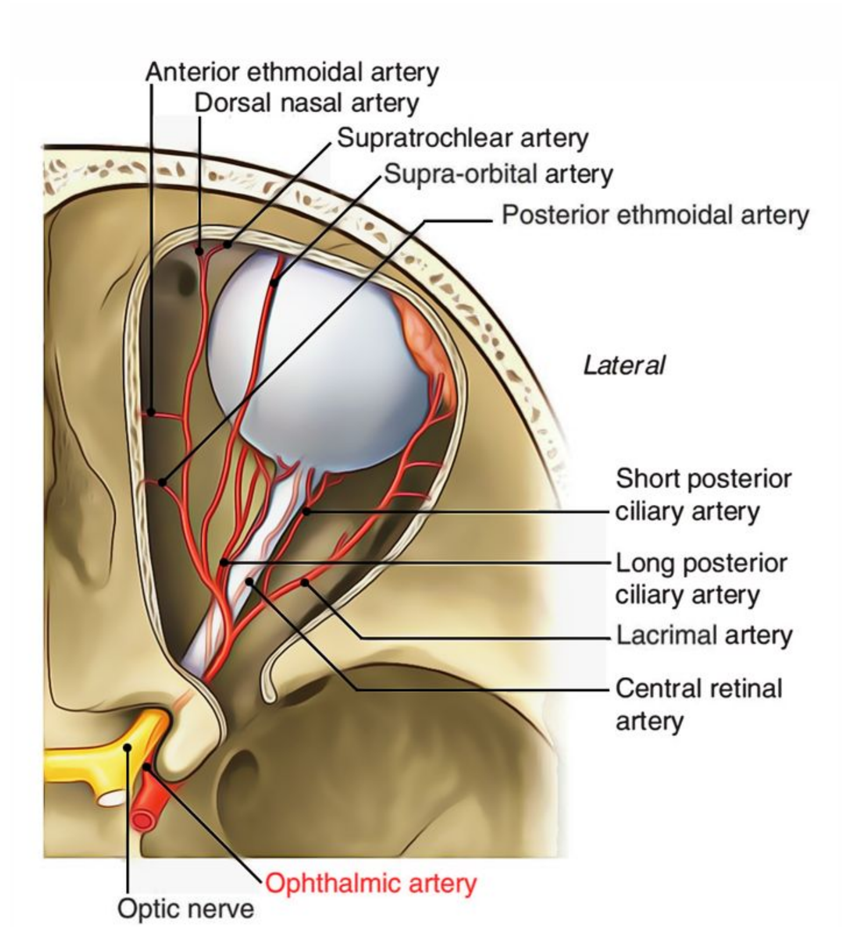
- It is a branch of internal carotid artery which arises close to the optic canal.
- It enters the orbit via optic canal.
- First it lies inferior and lateral to the optic nerve and then crosses superior to the nerve and proceeds anteriorly on the medial side.



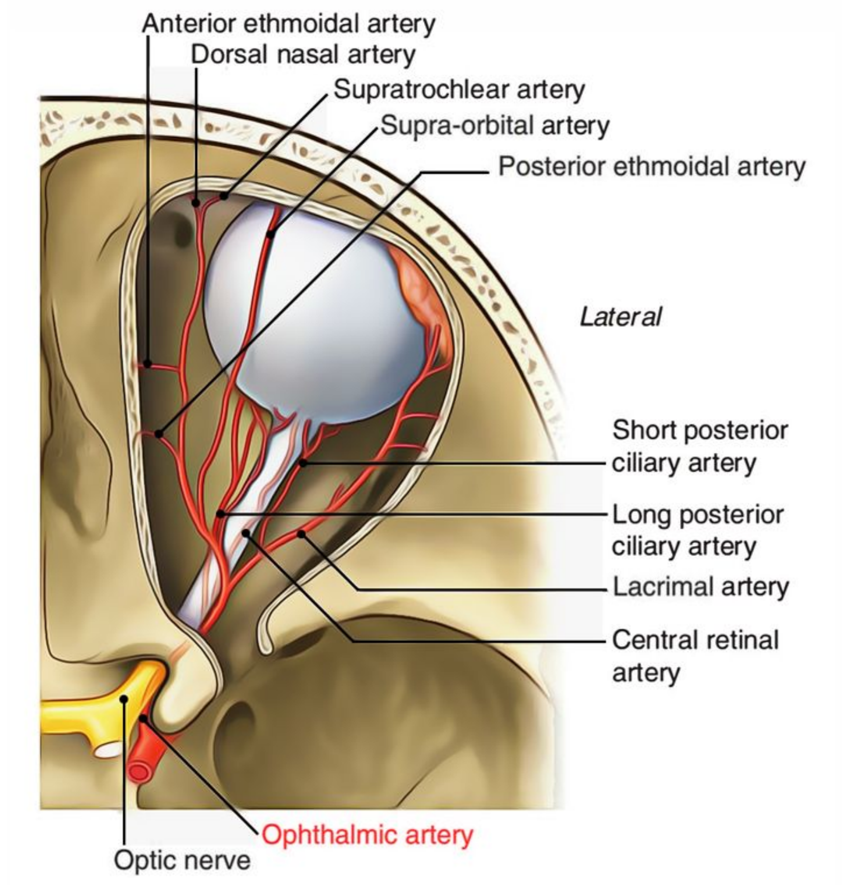
Branches of Ophthalmic Artery:

In the orbit, the ophthalmic artery gives off numerous branches as follows:

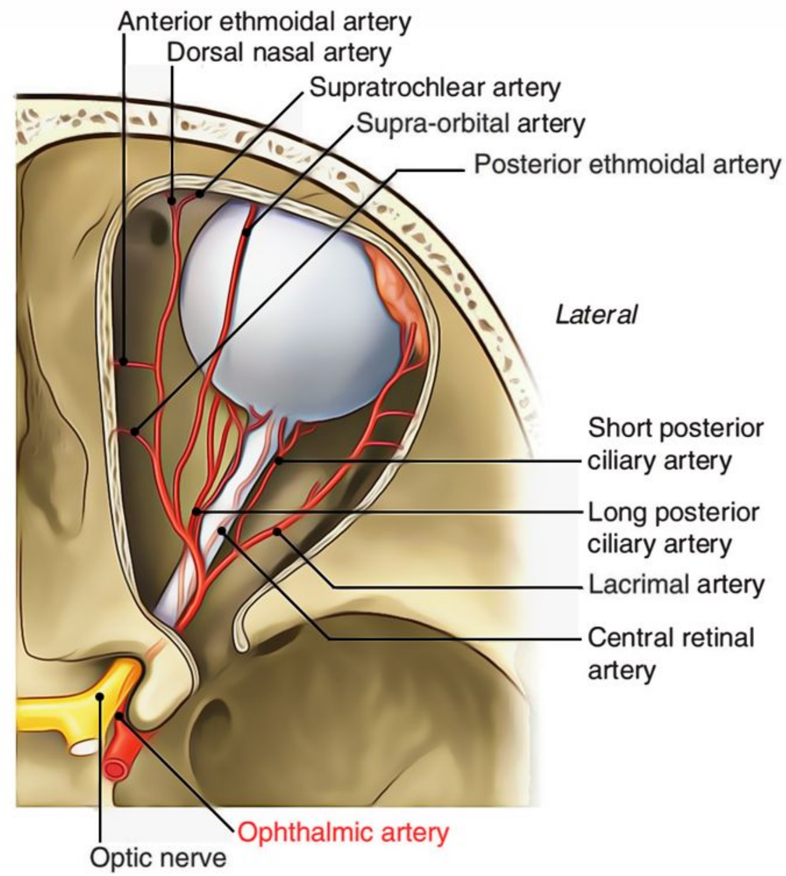
1. **Lacrimal artery:** arises laterally & supplies the lacrimal gland, the eyeball, & lateral side of the eyelid.
2. **Central retinal artery:** an important branch of the ophthalmic artery, it enters the optic nerve, proceeds down to its center to reach the retina. It is an end-artery, occlusion of this vessel may cause blindness.
3. **Long and short ciliary arteries:** pierce the sclera & supply the structures inside the eyeball.



4. **Muscular branches:** supply the intrinsic muscles of the eyeball.
5. **Supraorbital artery:** arises from the ophthalmic artery soon after it crosses the optic nerve, exits the orbit via supraorbital foramen & supplies the skin of the forehead & scalp up to vertex of the skull.
6. **Posterior ethmoidal artery:** exits the orbit via posterior ethmoidal foramen to supply the ethmoidal air cells & nasal cavity.
7. **Anterior ethmoidal artery:** exits the orbit through anterior ethmoidal foramen to supply the nasal septum & lateral wall & continues as dorsal nasal artery.



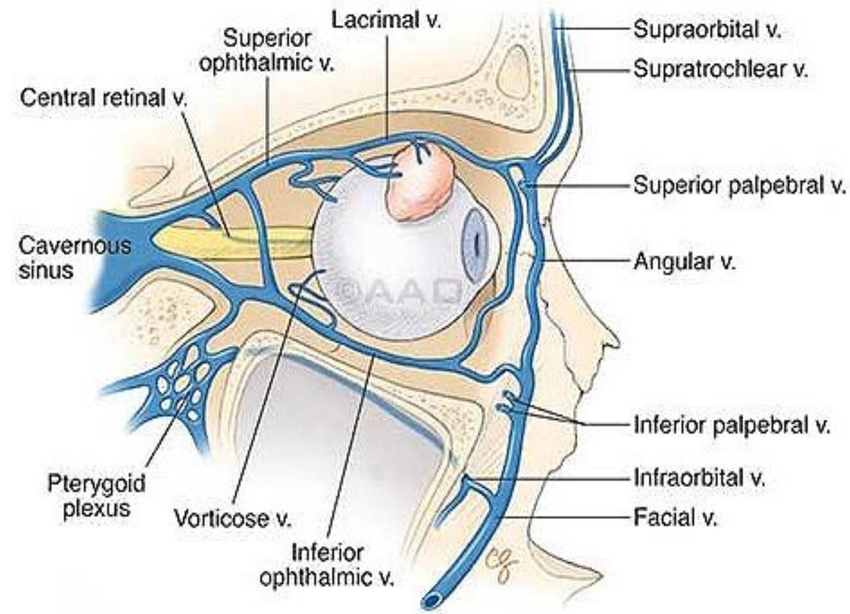
8. **Medial palpebral arteries:** supply the medial area of the upper & lower eyelids.
9. **Dorsal nasal artery:** one of the terminal branches of ophthalmic artery, supplies the upper surface of the nose.
10. **Supra-trochlear artery:** another terminal branch of ophthalmic artery, exits the orbit with supratrochlear nerve, & supplies the skin of the forehead & scalp up to vertex of the skull.



Ophthalmic Veins:

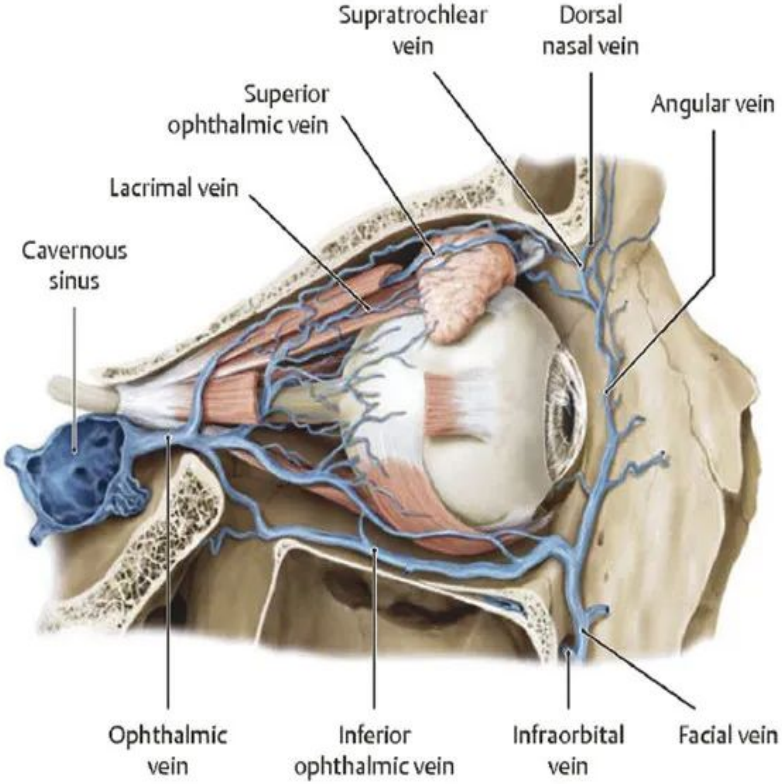
Superior Ophthalmic Vein:

- It accompanies the ophthalmic artery.
- Lies above the optic nerve.
- Tributaries correspond to the branches of the artery.
- Passes through the superior orbital fissure.
- Communicates anteriorly with the supraorbital and angular veins.



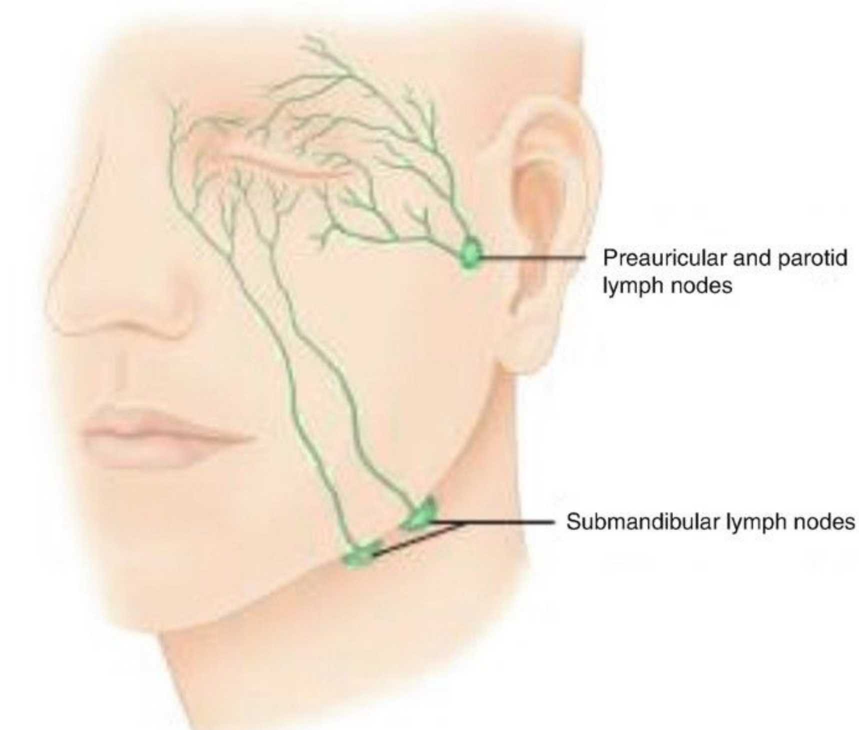
Inferior Ophthalmic Vein:

- Lies below the optic nerve.
- Receives tributaries from the lacrimal sac, the lower orbital muscles, & the eyelids.
- Ends either by joining the superior ophthalmic vein or drains directly into the cavernous sinus.



Lymphatics of the Orbit:

- The lymphatics drain into the preauricular/parotid lymphnodes.



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